

USER'S MANUAL MPD-200A

TAKASAGO ELECTRIC, INC.

1. Overview

MPD-200A is a compact, thin, and lightweight driver board that is specifically designed to drive Takasago piezoelectric micro pumps. It converts a single 5 V power and generates drive voltages as high as 250Vp-p at frequency of around 40 Hz. MPD-200A takes external signals for complete control of the pump's drive voltage and drive frequency. It also can operate stand-alone and drive a piezoelectric micro pump at 250 Vp-p and 40 Hz. MPD-200A enables quick and easy integration of Takasago piezoelectric micro pumps into your application settings.

The major features include:

- Compact, thin, and lightweight high-voltage generator circuit.
- Accepts external signals to control drive voltage and frequency.
- Also operates stand-alone at fixed voltage and frequency.
- Low power consumption.



2. General Safety Instructions

Use the following safety guidelines to help ensure your own personal safety and to help protect your equipment and working environment from potential damage.

	WARNING	Failure to observe instructions given with this symbol may result in injury or death by fire or electric shock.
	CAUTION	Failure to observe instructions given with this symbol may result in injury or damage to property damage by electric shock or other reason.

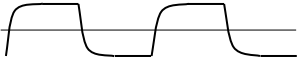
	Mark for "WARNING" and "CAUTION"		Mark for prohibition.		Mark for instruction.
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	WARNING		
	Do not touch high voltage parts. It may result in electric shock.		Turn off the switch or remove the power plug from the outlet immediately in event of abnormality. Failure to do so may result in fire or electric shock.
	Do not use in a wet environment. It may result in electric shock.		

	CAUTION			
	Turn off the power when not in use for long periods or when performing maintenance. Failure to do so may result in fire or electric shock.		Do not store in extremely humid, dusty or oily places or near the heat producing equipment. Doing so may result in malfunction, fire or electric shock.	

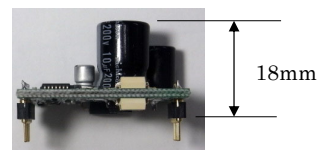
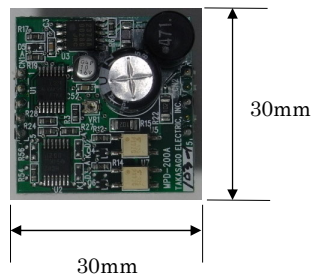
3. Specification

3-1) Features Summary

MPD-200A	
Input Voltage	5 VDC $\pm$ 5 %
Output Frequency	1 Hz to 60 Hz Varies with signal input at F in
Output Voltage	50 Vp-p to 340 Vp-p Varies with voltage input at V in
Drive Waveform	+ GND -  (Takasago Standard Wave)
Can be used with	SDMP302 / 306 / 320
Max. number of pumps	2
Power Consumption	Approx. 40 mW (Typical) Stand-by (SDMP302 x 1, at 60 Vp-p, Disabled) Approx. 85 mW (Typical) Stand-by (SDMP302 x 1, at 250 Vp-p, Disabled) Approx. 240 mW (Typical) In operation (SDMP302 x 1, at 250 Vp-p, 40 Hz)
Input Terminals	1. VDD Input voltage (DC 5 V) 2. V in Pump drive voltage input (0.2 to 4.5 V) 3. Enable Pump drive signal ON/OFF 4. F in Pump drive signal input (1 to 60 Hz) 5. GND GND
Output Terminals	1. High Voltage Out (-) Pump drive voltage output (-) 2. NC No connection 3. GND GND 4. NC No connection 5. High Voltage Out (+) Pump drive voltage output (+)
Internal Jumper Resistors	R51: Output Voltage Varies with Vin / Output voltage fixed (250 Vp-p approximately) R52: Output Frequency Varies with Fin / Output frequency fixed (40 Hz approximately)
Operating Temperature	-10 °C to + 60 °C
Storage Temperature	-20 °C to + 70 °C
Dimensions	30 x 30 x 18 mm When mounted on PCB
Mass	Approx. 9 g Including connector

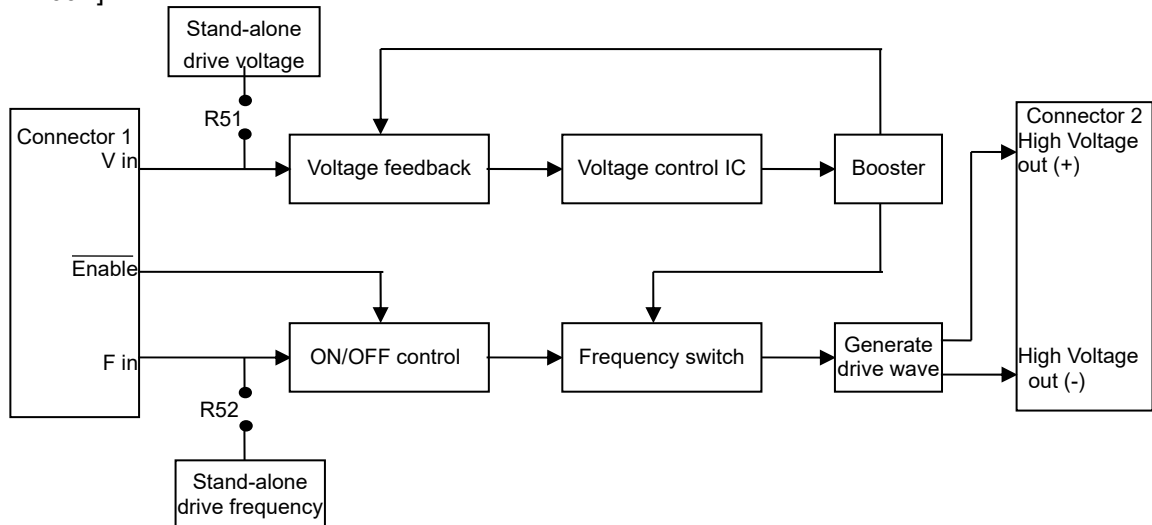
*) MPD-200A drives Takasago piezoelectric micro pumps with Takasago Standard waveform. Drive by Takasago Standard waveform provides higher flow rate and pressure performance than the sine wave drive does.

3-2) Appearance



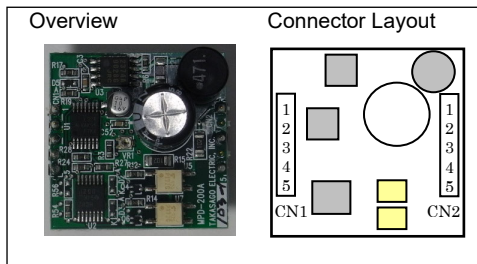
4. Circuitry Diagram

[MPD-200A]



5. Input/Output Terminals

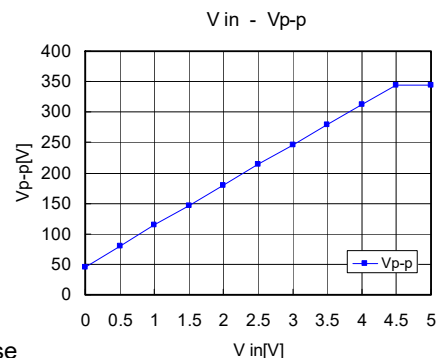
5-1) Layout

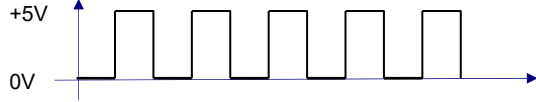


Names of parts	Function
Connector CN1	Input
Connector CN2	Output

5-2) Input Terminals

Terminal	Description
1. VDD	<Power Source> 5 VDC $\pm$ 5 %. Use a power source with little voltage fluctuation.
2. V in	<Voltage Input> Input range 0.2~4.5 V. The input of 0.2 V provides pump drive output of approx. 50 Vp-p, 3.0 V approx. 250 Vp-p, and 4.5 V approx. 340 Vp-p. Diagram to the right shows the relationship between V_{in} and the output drive voltage (Vp-p). Connecting more than one pump makes the output voltage a little lower than shown here. When an external control input is used, always verify the actual drive voltage by use of an oscilloscope. Always use drive voltage within range of the pump specification. Use of a voltage beyond that may affect reliability of the pump. Apply precautionary measures against electric noise contamination, such as minimization of the lead length and/or shielding of the leads. Noise contamination in V_{in} signal degrades stability of output voltage.



3. Enable	<Output ON/OFF> Drive OFF at logic high =5 V Drive ON at logic low =0 V This is a pull-down circuitry. No connection means Low and thus Pump Drive Output ON.
4. F in	<Frequency Input> Feed a square wave signal between 1 Hz and 60 Hz to generate a proper drive voltage. The square wave should have a logic Low at 0V and a logic High at 5V.  Signal input by a function generator or equivalent needs extra caution. Use +2.5V offset and set Low at 0V and High at 5V. The absence of this offset would inflict a negative voltage and may damage the driver board. Always feed square waves only. Triangular or sinusoidal waves are not recommended. Apply precautionary measures against electric noise contamination, such as minimization of the lead length and/or shielding of the leads. Noise contamination in Vin signal degrades stability of output voltage.
5. GND	Power GND. Connect VDD GND to this.

Round pin header (Male-Male).

Round pin IC socket (2.54 mm pitch, single 5 pins) is recommended.

5-3) Output Terminals

Terminal	Description
1. High Voltage Out (-)	Pump drive output (-) (connect to Black cable)
2. NC	No connection
3. GND	GND (No connection required)
4. NC	No connection
5. High Voltage Out (+)	Pump drive output (+) (connect to Red cable)

Round pin header (Male-Male).

Round pin IC socket (2.54 mm pitch, single 5 pins) is recommended.

5-4) Jumper Resistors

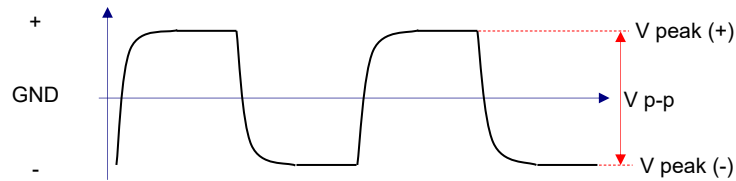
Jumper Resistor	OPEN (Factory default)	CLOSED
R51: Output Voltage	Varies with input at V in.	Fixed (approx. 250 Vp-p)
R52: Output Frequency	Varies with input at F in.	Fixed (approx. 40 Hz)

MPD-200A can provide drive output without taking external control signals. Shortcircuit the jumper resistors R51 and R52, and you have a drive output with fixed voltage/frequency. When you do use external control signals, however, always make sure these jumper resistors are open. A short circuit between R51 and R52 would disturb external control inputs to pin #2 (V in) and pin #4 (F in). When you select to close R51 for fixed voltage operation, please make sure your power source is stable and reasonably accurate at 5 V. Many commercial AC adaptors put out a higher voltage than their specification. A VDD voltage higher than 5 V will produce a lower output voltage than 250 V.

6. Waveforms and Flow Characteristics

6-1) Drive Signal Waveform

MPD-200A drives Takasago piezoelectric micro pumps with Takasago standard waveform.



The Pump Drive Output Voltage, V_{p-p} , means the distance between $V_{peak(+)}$ and $V_{peak(-)}$ in the diagram above. It is “the height between the hill top and the valley bottom of the output waveform.” Connect the GND terminal of your oscilloscope to the pin #1 High Voltage Out (-) of the MPD-200A, and the scope input terminal to the pin #5 High Voltage Out (+) and verify the actual waveform.

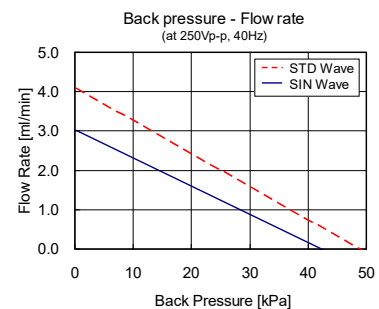
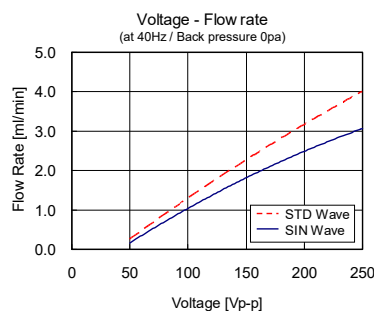
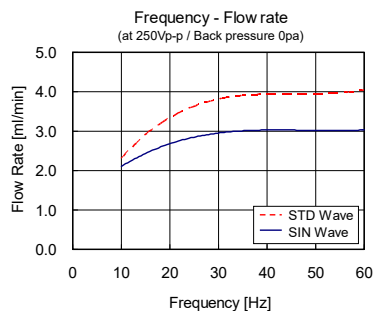
Your oscilloscope's GND terminal on another input channel must stay open. Do not connect them to MPD-200A GND (Input pin #5, or Output pin #3). This would eclipse half of the waveform, and your oscilloscope would show only a half of the picture.

Always use an oscilloscope to obtain correct V_{p-p} measurements. Multimeters, voltage meters, and similar instruments do not give correct measurements. Note that this is not a sinusoidal wave. Voltage meter readout $\times \sqrt{2} \times 2$ would not produce an accurate V_{p-p} value.

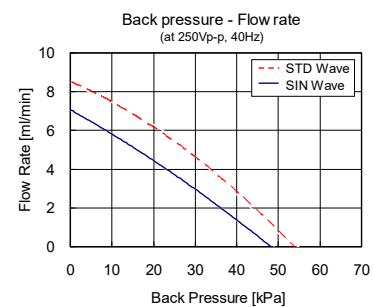
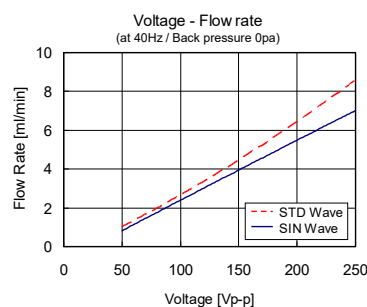
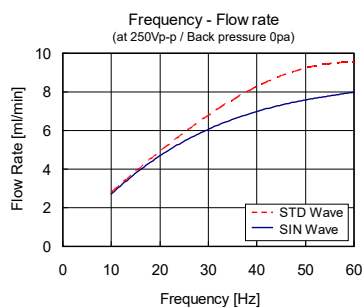
6-2) Flow Characteristics

The characteristics data in our related sales documents and this user's manual are based on sine wave driven operation. The flow rate and pump pressure would be larger if driven by Takasago standard wave, as it has more energy than the sine wave.

SDMP302	Sine (250 Vp-p, 40 Hz)	Takasago Standard (250 Vp-p, 40 Hz)
Maximum Flow Rate	3 ml/min. (approximately)	4 ml/min. (approximately)
Max. Discharge Pressure	40 kPa (approximately)	45 kPa (approximately)



SDMP306	Sine (250 Vp-p, 40 Hz)	Takasago Standard (250 Vp-p, 40 Hz)
Maximum Flow Rate	7 ml/min. (approximately)	8 ml/min (approximately)
Max. Discharge Pressure	45 kPa (approximately)	52 kPa (approximately)

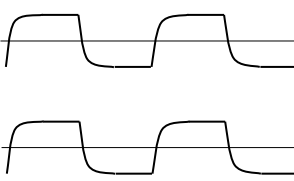
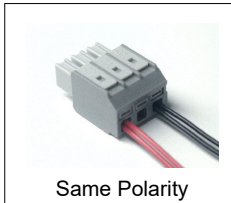
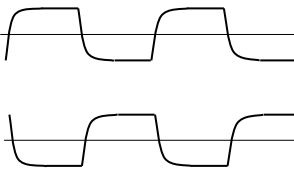
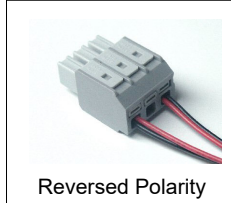


7. Using Multiple Pumps

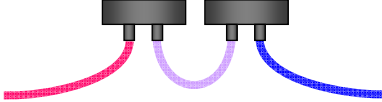
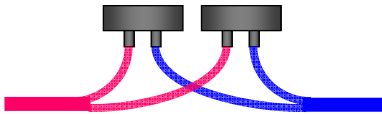
MPD-200A can drive up to 2 Takasago piezoelectric micro pumps at a time. Actuation timing of the Takasago piezoelectric micro pumps varies with polarity of the drive cables. When more than one pump is connected, their combined flow rate and pressure profile vary with the connection arrangement. Try various combinations of drive voltage/frequency and connection patterns to choose the best design for your application.

Connecting more than one pump affects the effective capacitance to the drive circuit's output stage. When two pumps are connected, the drive voltage falls a little bit, especially for higher range of output voltage and frequency. Also, the drive waveform tends to be more rounded off. Always check the actual waveform when two pumps are connected.

7-1) Drive Phase Combination

Phases	Operation timing
Same	Connect the red cable of pump A to the red cable of pump B, and the black of A to the black of B. The pumps suck and discharge at the same timing. Pump A  Pump B    Same Polarity The pumps may not operate at exactly the same timing, as response to drive input varies from particular pump to pump.
Reverse	Connect the red cable of pump A to the black cable of pump B, and the black of A to the red of B. The pumps suck and discharge at the reverse timing. Pump A  Pump B    Reversed Polarity The pumps may not operate at exactly the same timing, as response to drive input varies from particular pump to pump.

7-2) Connection Pattern

Pattern	Performance
Series	The combined flow rate of series-connected multiple pumps is virtually the same as that of one pump. The series connection of more than one pump improves the pressure withstand. The withstand is roughly proportional to the number of pumps connected.  The flow rate does not vary much with drive phase combination (same or reverse). The flow ripple is smaller with the reverse phase combination.
Parallel	The combined flow rate of parallel-connected multiple pumps is roughly proportional to the number of pumps connected. The pressure withstand is virtually the same as that of one pump.  The flow rate does not vary much with drive phase combination (same or reverse). The flow ripple is smaller with the reverse phase combination. Make sure that you use sufficiently thick tubings to accommodate the merged flow. Tubings where flows are joined should have cross section area proportional to the number of pumps involved, so that you can full enjoy the improved flow rate.

8. Changing Fixed Output Settings

MPD-200A can provide drive voltage output without taking external control signals. Shortcircuit both of R51 and R52 jumper resistors, and you have an output signal with fixed voltage and frequency, approx. at 250Vp-p and 40Hz. These values can be changed by replacing the components. For further details, please contact us.

9. Precautions

- MPD-200A is designed for incorporation to the system. Please incorporate the driver into the system for your safety.
- MPD-200A piezoelectric micro pumps driver board operates at a high voltage. Avoid contact with the circuitry board, electric parts, or connection terminals while the board is in operation, as this may cause injury.
- Piezoelectric elements tends to store electric charge just as electric capacitors do. Use caution when disconnecting the pump from the MPD-200A as discharge may cause injury. Excessive surge current, by short circuit or other repeated discharges, may affect reliability of the pump.
- MPD-200A is capable of generating 340 Vp-p at its maximum. This is higher than the upper limit that Takasago piezoelectric micro pumps can accept. Always make sure you use your piezoelectric micro pumps within their specifications. Voltages beyond their specification may degrade the pump's reliability. See the chart here for each pump's maximum voltage intake.
- Do not alter settings of the resistor parts on the circuitry board. They are adjusted to perform their optimum before leaving the Takasago facility.
- Do not feed negative signals to F in of MPD-200A. This may damage the circuitry of MPD-200A.
- Modification or remodelling of the driverboard invalidates our warrantee.

Pump Model	Maximum Vp-p
SDMP302	250Vp-p
SDMP306	250Vp-p
SDMP320	250Vp-p

10. Reminders

- **This is a prototype. It may or may not become commercially available with the specifications described in this document.**
- Please note that this is **not a RoHS compliant product** at this moment.
- The data shown here are typical values obtained in Takasago' laboratory tests, and are provided here for general information purposes only. Takasago disclaims any warranties, expressed or implied, as to this information and assumes no obligation or liability therefore.
- For use of this product in your application meant for export, please make sure you satisfy all the national and international legal and regulatory requirements.
- Specifications and appearance are subject to change without notice.
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11. Contact

If you have any questions, please feel free to contact us at any time.

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12. Revision History

Revision Number	Date	Revision Page	Revision Summary
1.00	2009/05/07	—	First Edition
1.01	2009/12/14	All pages All pages p.3-5 p.2-4 p.8 p.8	Deleted "CONFIDENTIAL" Revised the model number Revised the part number of the jumper Revised the photo, the dimension and the weight Revised "Changing Fixed Output Settings" in Chapter. 8 Added precautions (power supply and modification) in Chapter. 9
1.02	2011/06/01	p.1, 8 p.2, 8 p.5 p.8	Deleted 'prototype' in front page and Chapter. 10 Added 'Incorporation to the system' in Chapter.2 and 9 Added information about warranty in Chapter. 5 Revised inline connection in Chapter. 8
1.03	2015/09/01	p.1	Modifying the front page of the title

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